

Single-Channel Pipette with Fixed Gripper

Automated System

Project Overview:

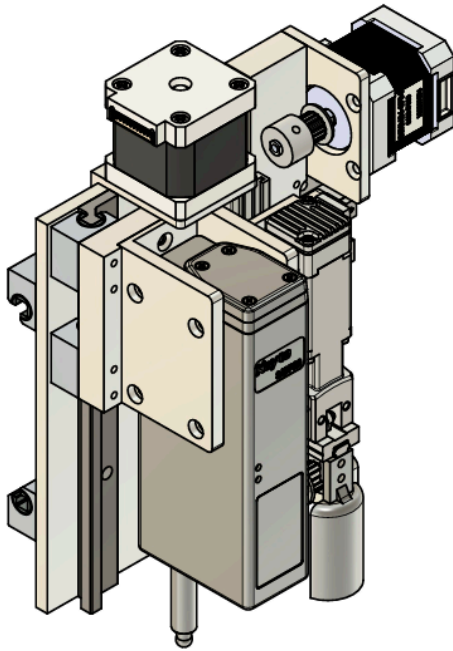
This project adapts a previously developed 8-channel dual-axis gantry system into a compact single-channel automated pipetting device with a fixed gripper. The system is designed to open individual test tubes (or capped sample tubes) using a fixed gripper, then perform precise single-channel pipetting of sample volume into a destination vessel. The mechanical components, materials, motors and control strategy follow the same design philosophy and parts selection as the original dual-axis 8-channel gantry system.

Technical Specifications:

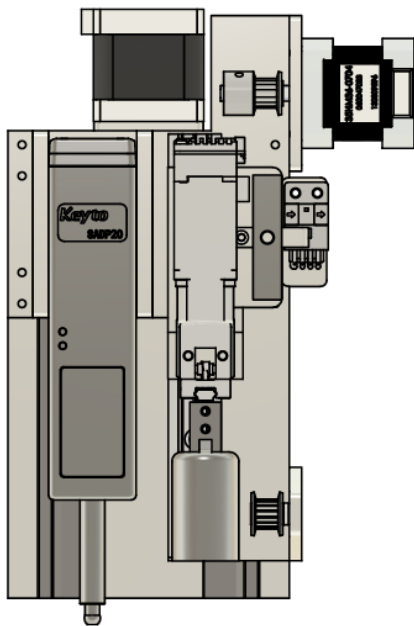
- **Vertical Motion:** Moons NEMA 17 Stepper Motor
- **Vertical Travel Mechanism:** Supported on single Igus slider for precise linear motion
- **Gripper:** Fixed type, actuated using a mini stepper/DC motor with cam-based rotation
- **Pipette:** Single-channel (user-defined model)
- **Frame Material:** Aluminum (lightweight & rigid)
- **Travel Range** - Vertical: 200 mm, Horizontal: Can be adjusted
- **Couplers / Attachments:** Custom machined for accurate alignment
- **Maintenance:** Easy replacement and upgrade of mechanical parts
- **Drive Type:** V-belt driven mechanism for controlled motion

CAD Representation:

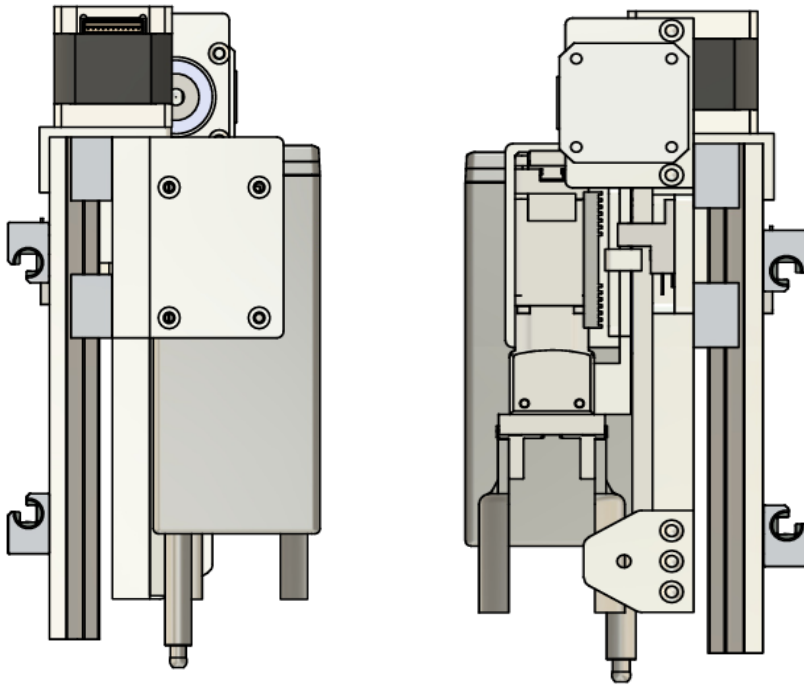
- **Isometric View:** Complete assembly showing fixed gripper and pipette system.



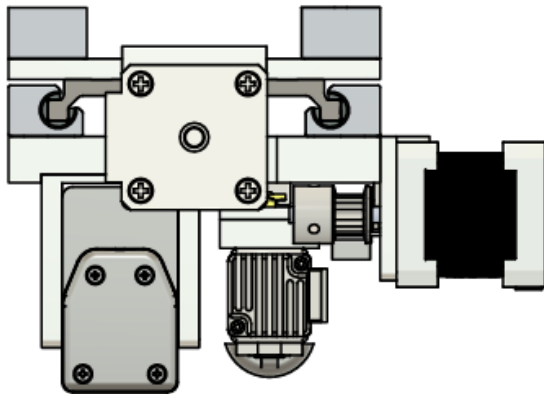
- **Front View:** Vertical motion of the pipette along Z-axis.



- **Side View:** Relationship between pipette and fixed gripper.



- Top View: Layout showing base plate and fixed gripper position.



Design Highlights:

- Vertical Motion:

- Single stepper motor drives pipette smoothly.

- Igus slider ensures low-friction and precise movement.
- V-belt transmission for smooth and reliable motion.
- **Fixed Gripper:**
 - Actuated using a small motor for cap rotation or removal.
 - Provides stable hold for test tubes during pipetting.
- **Modularity & Maintenance:**
 - Components can be replaced or upgraded easily.
 - Can be integrated into multi-channel systems in the future.
- **Performance:**
 - High repeatability and accuracy for biomedical applications.

Materials & Components:

- Stepper Motor: Moons NEMA 17 dual shaft
- Linear Guides: Igus sliders for smooth motion
- Frame: Aluminum for rigidity and light weight
- Pipette: Single-channel
- Gripper: Fixed, cam-actuated
- Couplers / Attachments: Custom machined for precise fit
- Drive Mechanism: V-belt system for controlled motion

Challenges Faced & Solutions:

- **Challenge:** Maintaining smooth vertical motion with minimal vibration.
Solution: Used high-quality Igus slider and optimized alignment.
- **Challenge:** Reliable gripper action across different tube types.
Solution: Designed adjustable cam-based gripper with flexible cap holders.
- **Challenge:** Ensuring accurate pipette positioning.
Solution: Used V-belt drive with microstepping control.
- **Challenge:** Compact integration of gripper and pipette.
Solution: Optimized base layout for minimum footprint.
- **Challenge:** Preventing cross-contamination.
Solution: Integrated disposable pipette tip handling and tip-touch function.

